

From Transistor Bias to Amplifier Gain: Large-Signal and Small-Signal Models of a Common-Emitter Stage

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1 The two questions we ask about an amplifier

A transistor amplifier has a resting state and a response to a changing input. We analyze these separately:

1. **Large-signal (DC) analysis:** Where is the Q-point? What happens if the transistor reaches cutoff or saturation?
2. **Small-signal (AC) analysis:** For a sufficiently small change around the Q-point, what are the voltage gain, input resistance, and output resistance?

The order matters. We first find the Q-point, because the parameters of the small-signal model depend on the current at that point.

Intuition. A curved road has a different local slope at different locations. The transistor's large-signal characteristic is the curved road; bias selects our location; the small-signal model uses the slope of the curve near that location.

Throughout, we consider an NPN common-emitter (CE) stage. Unless stated otherwise, numerical examples use

$$V_{CC} = 12 \text{ V}, \quad R_C = 1 \text{ k}\Omega, \quad \beta = 100.$$

The value $\beta = 100$ is an illustrative assumption, not a guaranteed property of a particular transistor.

2 Large-signal transistor model

2.1 Operating regions

A useful first model distinguishes three regions:

Region	Junction conditions	Circuit-level behavior
Cutoff	Base-emitter junction not appreciably forward biased	I_C is approximately zero.
Forward-active	Base-emitter forward biased; base-collector reverse biased	Collector current responds strongly to V_{BE} ; useful region for linear amplification.
Saturation	Both junctions forward biased	V_{CE} is small; the collector circuit limits the current.

A basic piecewise approximation uses $V_{BE} \approx 0.7$ V when the transistor conducts and $V_{CE(\text{sat})} \approx 0.2$ V in saturation. These numbers are convenient estimates, not universal constants.

2.2 A more informative forward-active model

In forward-active operation, collector current depends approximately exponentially on base-emitter voltage:

$$I_C \approx I_S e^{V_{BE}/V_T}, \quad (1)$$

where I_S is a transistor-dependent scale current and $V_T = kT/q$ is the thermal voltage. Near room temperature, $V_T \approx 25$ – 26 mV.

Equation (1) explains two important observations:

- A modest change in V_{BE} can produce a substantial change in collector current.
- $V_{BE} = 0.7$ V is only an approximate design shortcut; there is no perfectly sharp “turn-on” voltage.

We also often write

$$I_C \approx \beta I_B, \quad I_E = I_C + I_B.$$

These are useful current relations in forward-active operation. They should not be applied blindly in cutoff or saturation.

3 Large-signal analysis finds the Q-point

3.1 Grounded-emitter example

Suppose a bias network establishes $I_{CQ} = 6$ mA. With the emitter grounded,

$$V_{CEQ} = V_{CC} - I_{CQ}R_C = 12 - (6 \text{ mA})(1 \text{ k}\Omega) = 6 \text{ V}.$$

Thus

$$Q: \quad I_{CQ} = 6 \text{ mA}, \quad V_{CEQ} = 6 \text{ V}.$$

The approximate base current is $I_{BQ} = I_{CQ}/\beta = 60$ μ A.

The load-line limits are approximately $I_C = 0$ at cutoff and $V_{CE} \approx 0.2$ V at saturation. The chosen Q-point leaves room for signal movement toward both limits.

3.2 Example with an emitter resistor

Now add $R_E = 500$ Ω between emitter and ground. Suppose a suitable bias network holds the base near 2.7 V. Using $V_{BE} \approx 0.7$ V,

$$\begin{aligned} V_{EQ} &\approx V_{BQ} - V_{BEQ} = 2.0 \text{ V}, \\ I_{EQ} &\approx \frac{V_{EQ}}{R_E} = \frac{2.0}{500} = 4.0 \text{ mA}. \end{aligned}$$

Since $I_E = I_C(1 + 1/\beta)$, a calculation retaining $\beta = 100$ gives

$$I_{CQ} = \frac{\beta}{\beta + 1} I_{EQ} \approx 3.96 \text{ mA}.$$

For quick calculations we will use $I_{CQ} \approx 4 \text{ mA}$. Then

$$\begin{aligned} V_{CQ} &\approx 12 - (4 \text{ mA})(1 \text{ k}\Omega) = 8 \text{ V}, \\ V_{EQ} &\approx 2 \text{ V}, \\ V_{CEQ} &= V_{CQ} - V_{EQ} \approx 6 \text{ V}. \end{aligned}$$

Therefore,

$$\boxed{Q: \quad I_{CQ} \approx 4 \text{ mA}, \quad V_{CEQ} \approx 6 \text{ V}.}$$

Do not confuse V_C with V_{CE} . With an emitter resistor, the emitter is above ground. In this example $V_C \approx 8 \text{ V}$, $V_E \approx 2 \text{ V}$, and $V_{CE} \approx 6 \text{ V}$.

4 From the large-signal curve to the small-signal model

Let the base-emitter voltage and collector current change slightly around their Q-point values:

$$v_{BE}(t) = V_{BEQ} + v_{be}(t), \quad i_C(t) = I_{CQ} + i_c(t).$$

Here uppercase V_{BEQ} and I_{CQ} denote resting values. Lowercase v_{be} and i_c denote *small changes* around those values.

Substitute $V_{BEQ} + v_{be}$ into Equation (1):

$$I_{CQ} + i_c \approx I_S e^{(V_{BEQ} + v_{be})/V_T} = I_{CQ} e^{v_{be}/V_T}.$$

For $|v_{be}| \ll V_T$, use $e^x \approx 1 + x$:

$$I_{CQ} + i_c \approx I_{CQ} \left(1 + \frac{v_{be}}{V_T} \right).$$

Subtract I_{CQ} from both sides:

$$\boxed{i_c \approx g_m v_{be}, \quad g_m = \frac{I_{CQ}}{V_T}.} \quad (2)$$

The quantity g_m , called *transconductance*, tells us how much collector current changes per volt of small-signal base-emitter voltage.

Significance of the Q-point. Because $g_m = I_{CQ}/V_T$, changing the bias current changes the small-signal gain. A transistor does not have one fixed voltage gain independent of its bias and surrounding circuit.

5 Why does the hybrid- π model have these elements?

The hybrid- π model is a local description of what happens when we make *small changes around the Q-point*. It is not a claim that three separate physical components are hidden inside the transistor. Each circuit element represents one measured relationship between small changes in voltage and current.

5.1 The base-emitter resistance r_π

At the Q-point, the base draws a DC current I_{BQ} . If we raise the base-emitter voltage slightly, the base current increases. Over a sufficiently small range, we can describe the change by

$$i_b \approx \frac{v_{be}}{r_\pi}, \quad \text{where} \quad r_\pi \equiv \left. \frac{\partial V_{BE}}{\partial I_B} \right|_Q.$$

This is why we draw r_π *between base and emitter*: a small voltage across those terminals causes a small base current.

In forward-active operation, the small changes also satisfy $i_c \approx \beta i_b$ and $i_c = g_m v_{be}$. Combining them,

$$g_m v_{be} \approx \beta \frac{v_{be}}{r_\pi},$$

so

$$r_\pi = \frac{\beta}{g_m} = \frac{\beta V_T}{I_{CQ}}.$$

Intuition. r_π is the resistance the signal approximately “sees” when looking into the transistor’s base with the emitter held at AC ground. It is a *differential* resistance: the ratio of a small change in voltage to a small change in current. It is not V_{BEQ}/I_{BQ} .

5.2 The controlled current source $g_m v_{be}$

In forward-active operation, collector current depends strongly on base-emitter voltage:

$$I_C \approx I_S e^{V_{BE}/V_T}.$$

The slope of this curve at the Q-point is

$$\left. \frac{\partial I_C}{\partial V_{BE}} \right|_Q = \frac{I_{CQ}}{V_T} \equiv g_m.$$

Hence a small change v_{be} produces approximately

$$i_c = g_m v_{be}.$$

We represent this relation by a *voltage-controlled current source* from collector to emitter. It is called “controlled” because its current is set by the voltage v_{be} elsewhere in the model.

Intuition. The small input voltage v_{be} controls a larger change in collector current. That current flowing through the collector load creates the output-voltage change. This is the central amplification mechanism of the CE stage.

5.3 Why an output resistance r_o ?

An idealized forward-active transistor would have collector current determined only by V_{BE} . If we held V_{BE} fixed and changed V_{CE} , its collector current would remain exactly constant.

A real transistor behaves somewhat differently: increasing V_{CE} usually causes a small increase in I_C , even when V_{BE} is fixed. This is the *Early effect*. Around the Q-point, we describe that extra current change by

$$i_{c,\text{Early}} \approx \frac{v_{ce}}{r_o}, \quad r_o \equiv \left(\frac{\partial I_C}{\partial V_{CE}} \right)^{-1} \bigg|_Q.$$

Thus we draw a resistor r_o between collector and emitter. With the Early effect included, the incremental collector current is approximately

$$i_c \approx g_m v_{be} + \frac{v_{ce}}{r_o}.$$

A common rough estimate is

$$r_o \approx \frac{V_A}{I_{CQ}},$$

where V_A is the transistor's Early voltage. If the Early effect is neglected, r_o is treated as infinitely large and that branch can be removed.

Intuition. The controlled source describes how the *input* changes collector current. The r_o branch describes how the *collector voltage itself* slightly changes collector current. A large r_o means this second effect is weak.

5.4 Putting the three relationships together

The complete low-frequency hybrid- π model therefore describes three small-signal relationships:

$$\underbrace{i_b = \frac{v_{be}}{r_\pi}}_{\text{base input current}}, \quad \underbrace{i_c = g_m v_{be} + \frac{v_{ce}}{r_o}}_{\text{collector current}}.$$

For the earlier example with $I_{CQ} = 4$ mA, $\beta = 100$, $V_T = 26$ mV, and an illustrative $V_A = 100$ V,

$$g_m \approx 0.154 \text{ S}, \quad r_\pi \approx 650 \text{ } \Omega, \quad r_o \approx 25 \text{ k}\Omega.$$

For instance, a very small $v_{be} = 1$ mV produces, if v_{ce} is held fixed,

$$i_b \approx \frac{1 \text{ mV}}{650 \text{ } \Omega} \approx 1.54 \text{ } \mu\text{A}, \quad i_c \approx (0.154 \text{ S})(1 \text{ mV}) = 154 \text{ } \mu\text{A}.$$

Their ratio is approximately $100 = \beta$, as expected. If v_{ce} also changes by 1 V, the r_o branch contributes approximately another $1 \text{ V}/25 \text{ k}\Omega = 40 \text{ } \mu\text{A}$ to the incremental collector current.

These values apply *near the stated Q-point*. A large signal or a move into cutoff or saturation requires returning to the large-signal model.

6 The hybrid- π small-signal model

For a CE amplifier, the basic hybrid- π model contains:

1. A resistance r_π between base and emitter.
2. A controlled current source $g_m v_{be}$ from collector to emitter.
3. Optionally, an output resistance r_o between collector and emitter, accounting for the Early effect.

The main parameters are

$$\boxed{g_m = \frac{I_{CQ}}{V_T}, \quad r_\pi = \frac{\beta}{g_m} = \frac{\beta V_T}{I_{CQ}}, \quad r_o \approx \frac{V_A}{I_{CQ}}.} \quad (3)$$

Here V_A is the Early voltage, a device parameter. The r_o expression is an approximation; if V_A is not specified, we often begin with $r_o \rightarrow \infty$ and later assess whether neglecting it was reasonable.

Because $i_b = v_{be}/r_\pi$ and $i_c = g_m v_{be}$,

$$\frac{i_c}{i_b} = g_m r_\pi = \beta.$$

Thus the hybrid- π model is consistent with the familiar small-signal current gain β .

Numerical model at $I_{CQ} = 4 \text{ mA}$

Using $V_T = 26 \text{ mV}$ and $\beta = 100$,

$$g_m = \frac{4 \text{ mA}}{26 \text{ mV}} \approx 0.154 \text{ S},$$

$$r_\pi = \frac{100}{0.154} \approx 650 \text{ } \Omega.$$

If, illustratively, $V_A = 100 \text{ V}$, then

$$r_o \approx \frac{100 \text{ V}}{4 \text{ mA}} = 25 \text{ k}\Omega.$$

7 Rules for constructing the AC small-signal circuit

At the signal frequency of interest:

- Replace the transistor with its small-signal model.
- Replace an ideal DC voltage supply by AC ground, because its voltage has no signal variation.
- Treat a sufficiently large coupling or bypass capacitor as an AC short circuit.
- Retain resistors; their DC and AC roles may differ.
- Include any connected signal load R_L .

These capacitor approximations describe the *midband* range. At low frequencies, a capacitor may not behave like a short; at high frequencies, transistor capacitances also become important.

8 Voltage gain of a grounded-emitter CE amplifier

First neglect r_o . If the emitter is AC-grounded, then

$$v_{be} = v_b.$$

A small positive base voltage produces $i_c = g_m v_{be}$. This additional current through the effective collector load lowers the collector voltage. If an AC-coupled load R_L is present, let

$$R'_C = R_C \parallel R_L.$$

Then

$$v_o = -i_c R'_C = -g_m v_{be} R'_C.$$

The gain *from transistor base to collector* is therefore

$$\boxed{A_{v,b} \equiv \frac{v_o}{v_b} \approx -g_m (R_C \parallel R_L).} \quad (4)$$

Without a separate R_L , replace $R_C \parallel R_L$ by R_C . The minus sign means a positive change at the base produces a negative change at the collector.

If r_o matters, a first approximation for an AC-grounded emitter is

$$A_{v,b} \approx -g_m (R_C \parallel R_L \parallel r_o).$$

For the 4 mA example with $R_C = R_L = 1 \text{ k}\Omega$, neglecting r_o gives

$$A_{v,b} \approx -(0.154)(500) \approx -77.$$

This is a *local small-signal gain*. It does not mean a large input can be multiplied by 77 without clipping.

9 Emitter degeneration: an unbypassed R_E

Now retain $R_E = 500 \text{ }\Omega$ without an emitter-bypass capacitor. An increased base voltage raises emitter current, which raises emitter voltage. This opposes the increase in $v_{be} = v_b - v_e$.

Neglecting r_o , the base current i_b produces

$$v_{be} = i_b r_\pi, \quad i_e \approx (\beta + 1)i_b, \quad v_e \approx (\beta + 1)i_b R_E.$$

Consequently,

$$v_b = v_{be} + v_e = i_b [r_\pi + (\beta + 1)R_E].$$

The collector output is

$$v_o = -i_c (R_C \parallel R_L) = -\beta i_b (R_C \parallel R_L).$$

Divide the last expression by v_b :

$$\boxed{A_{v,b} \approx -\frac{\beta (R_C \parallel R_L)}{r_\pi + (\beta + 1)R_E}.} \quad (5)$$

Using $\beta = g_m r_\pi$, and approximating $(\beta + 1)/\beta \approx 1$ for large β , this can also be written

$$\boxed{A_{v,b} \approx -\frac{R_C \parallel R_L}{r_e + R_E}, \quad r_e \approx \frac{1}{g_m} = \frac{V_T}{I_{CQ}}.} \quad (6)$$

The r_e here is the transistor's *intrinsic small-signal emitter resistance*; it is not the external resistor R_E .

Numerical comparison

For $I_{CQ} \approx 4 \text{ mA}$,

$$r_e \approx \frac{26 \text{ mV}}{4 \text{ mA}} = 6.5 \Omega.$$

With $R_C = 1 \text{ k}\Omega$ and no external AC load,

$$A_{v,b} \approx -\frac{1000}{500 + 6.5} \approx -1.97.$$

Using the more exact β -based expression with $r_\pi \approx 650 \Omega$ gives

$$A_{v,b} \approx -\frac{100(1000)}{650 + 101(500)} \approx -1.96.$$

If $R_L = 1 \text{ k}\Omega$ is connected through a coupling capacitor, $R_C \parallel R_L = 500 \Omega$, and the gain is approximately -0.98 .

Trade-off. An unbypassed R_E reduces voltage gain, but the negative feedback makes gain less sensitive to variations in the transistor and improves linearity.

9.1 If the emitter resistor is bypassed

Suppose a capacitor is placed across R_E . For DC, that capacitor is open, so R_E still helps establish and stabilize the Q-point. At midband, if the capacitor effectively shorts R_E , the emitter is approximately at AC ground and the gain approaches Equation (4).

The transition is frequency-dependent: the emitter is not perfectly AC-grounded at every frequency.

10 Input resistance

Input resistance tells us how heavily the amplifier loads the circuit driving its input. At a chosen input port,

$$R_{\text{in}} = \frac{\text{small-signal voltage at the port}}{\text{small-signal current entering the port}}.$$

10.1 Looking into the transistor base

With the emitter AC-grounded,

$$R_{\text{in,base}} \approx r_\pi. \quad (7)$$

With an unbypassed R_E , the base must supply a small change in base-emitter voltage *and* a change in emitter voltage:

$$R_{\text{in,base}} \approx r_\pi + (\beta + 1)R_E. \quad (8)$$

For the 4 mA , $R_E = 500 \Omega$ example,

$$R_{\text{in,base}} \approx 650 + 101(500) = 51.15 \text{ k}\Omega.$$

Thus emitter degeneration greatly increases the resistance seen looking directly into the base.

10.2 Resistance seen at the amplifier input terminal

A practical bias divider also connects the base to AC ground through its two resistors. Denote their parallel combination by

$$R_B = R_1 \parallel R_2.$$

If the input coupling capacitor is an AC short, the source sees

$$\boxed{R_{\text{in}} \approx R_B \parallel R_{\text{in,base}}.} \quad (9)$$

For example, if $R_B = 20 \text{ k}\Omega$ and the unbypassed transistor presents $51.15 \text{ k}\Omega$,

$$R_{\text{in}} \approx 20 \text{ k}\Omega \parallel 51.15 \text{ k}\Omega \approx 14.4 \text{ k}\Omega.$$

Although R_E made the transistor's own base input resistance large, the bias network still limits the resistance seen by the source.

11 Source loading and overall gain

The gain formulas above use v_b , the voltage that actually reaches the base. A signal source may have its own resistance R_S . The source and amplifier input resistance form a voltage divider:

$$v_b = v_s \frac{R_{\text{in}}}{R_S + R_{\text{in}}}.$$

Therefore the overall gain from source voltage to output is

$$\boxed{\frac{v_o}{v_s} = \underbrace{\frac{v_o}{v_b}}_{\text{amplifier base-to-output gain}} \underbrace{\frac{R_{\text{in}}}{R_S + R_{\text{in}}}}_{\text{input loading}}.} \quad (10)$$

For example, take $A_{v,b} \approx -1.96$, $R_{\text{in}} \approx 14.4 \text{ k}\Omega$, and $R_S = 1 \text{ k}\Omega$. Then

$$\frac{v_b}{v_s} \approx \frac{14.4}{1 + 14.4} = 0.935, \quad \frac{v_o}{v_s} \approx -1.96(0.935) \approx -1.83.$$

This distinction explains why a measured gain can differ from the gain calculated between the transistor's base and collector.

12 Output resistance

Output resistance describes how much the output voltage changes when a load draws current. To find it, remove the external load R_L , set the independent input signal to zero, and look back into the output terminal using a small test voltage and current:

$$R_{\text{out}} = \frac{v_{\text{test}}}{i_{\text{test}}}.$$

At midband, the output coupling capacitor is treated as an AC short.

For an AC-grounded emitter, the simple hybrid- π approximation gives

$$\boxed{R_{\text{out}} \approx R_C \parallel r_o.} \quad (11)$$

If $r_o \gg R_C$, then $R_{\text{out}} \approx R_C$. For $R_C = 1 \text{ k}\Omega$ and $r_o = 25 \text{ k}\Omega$,

$$R_{\text{out}} \approx 1 \text{ k}\Omega \parallel 25 \text{ k}\Omega \approx 962 \Omega.$$

With an unbypassed emitter resistor, feedback generally increases the resistance looking into the *transistor collector* compared with r_o alone. The total stage output resistance still includes R_C in parallel, so $R_{\text{out}} \approx R_C$ is often a useful first estimate when the transistor's collector resistance is much larger than R_C . An exact calculation must also specify the source resistance and bias network seen at the base.

Why output resistance matters

An external load R_L draws signal current. A finite output resistance means some of the amplifier's available signal voltage is lost internally. In the elementary small-signal model, this is why the collector load in the gain formula becomes $R_C \parallel R_L$, rather than simply R_C .

13 Small-signal analysis has a limited range

The hybrid- π model is a *local linear approximation*. It works when the signal produces sufficiently small changes around the Q-point. In particular:

- The transistor should remain in forward-active operation.
- The small-signal parameters should not change greatly during the cycle.
- Coupling and bypass capacitor assumptions should be appropriate for the signal frequency.

For example, a gain of -77 predicts that a 1 mV base signal produces about 77 mV at the collector under its stated assumptions. It does *not* imply that a 0.5 V base signal can produce 38.5 V from a 12 V supply. Such a signal would leave the small-signal regime and encounter cutoff or saturation.

Use the two analyses together. The large-signal circuit determines the Q-point and clipping limits. The small-signal circuit predicts gain and resistances for small movements around that Q-point.

14 A compact analysis workflow

For a new CE amplifier, proceed in this order:

1. **Find the DC Q-point.** Treat capacitors as open. Calculate I_{CQ} , V_{BQ} , V_{EQ} , V_{CQ} , and V_{CEQ} . Check that the transistor is forward-active.
2. **Calculate model parameters.** Use $g_m = I_{CQ}/V_T$, $r_\pi = \beta/g_m$, and r_o if its device information is available.
3. **Draw the AC equivalent.** Set the DC supply to AC ground. Apply the appropriate capacitor approximations for the frequency of interest.
4. **Find base-to-output gain.** Include R_L , any unbypassed R_E , and r_o if needed.
5. **Find input and output resistances.** Include the bias network in the input resistance.
6. **Find source-to-output gain.** Account for the divider formed by R_S and R_{in} .
7. **Check signal swing.** Confirm that the predicted waveform remains within cutoff and saturation limits.

15 Key formulas at a glance

Quantity	Approximate expression
Transconductance	$g_m = I_{CQ}/V_T$
Base-emitter small-signal resistance	$r_\pi = \beta/g_m$
Grounded-emitter base-to-output gain	$A_{v,b} \approx -g_m(R_C \parallel R_L)$
Unbypassed- R_E base-to-output gain	$A_{v,b} \approx -\frac{\beta(R_C \parallel R_L)}{r_\pi + (\beta + 1)R_E}$
Input resistance looking into base	$r_\pi + (\beta + 1)R_E$
Input resistance including bias divider	$R_B \parallel [r_\pi + (\beta + 1)R_E]$
Output resistance, AC-grounded emitter	$R_C \parallel r_o$
Overall gain from a source with resistance R_S	$\frac{v_o}{v_s} = A_{v,b} \frac{R_{in}}{R_S + R_{in}}$